

Written Report on Discriminative vs Generative Models on MNIST Dataset

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1. Introduction

This report compares discriminative and generative modeling approaches using the MNIST handwritten digit dataset. The discriminative model (logistic regression) is extended with regularization and compared to non-linear classifiers. A generative adversarial network (GAN) is implemented to synthesize digit images. The goals include evaluating classification performance, regularization effects, training stability, and the utility of synthetic data.

2. Methods

Dataset

- **MNIST:** 28×28 grayscale images of digits 0–9.
- Training subset: 5,000 samples (for computational efficiency).
- Test set: 10,000 samples.
- Preprocessing: Images flattened and normalized to $[0,1]$ for discriminative models; normalized to $[-1,1]$ for GAN.

Discriminative Models

- **Logistic Regression:** Multinomial with L1/L2 regularization.
- **SVM:** Linear kernel.
- **Decision Tree:** Default scikit-learn parameters.
- **Evaluation:** 5-fold cross-validation; mean accuracy and standard deviation reported.

Generative Model

- **GAN Architecture:**
 - Generator: MLP with hidden layers [256, 512, 1024], outputting 28×28 images via Tanh.
 - Discriminator: MLP with hidden layers [512, 256], outputting binary classification via Sigmoid.
 - **Training:** Adam optimizer (LR=2e-4, $\beta_1=0.5$), BCE loss, label smoothing (0.9 for real, 0.0 for fake).
 - **Evaluation:** Sample quality, loss curves, mode collapse, and synthetic data utility.
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3. Results and Observations

Discriminative Models

Table 1: 5-Fold Cross-Validation Accuracy (Mean ± Std)

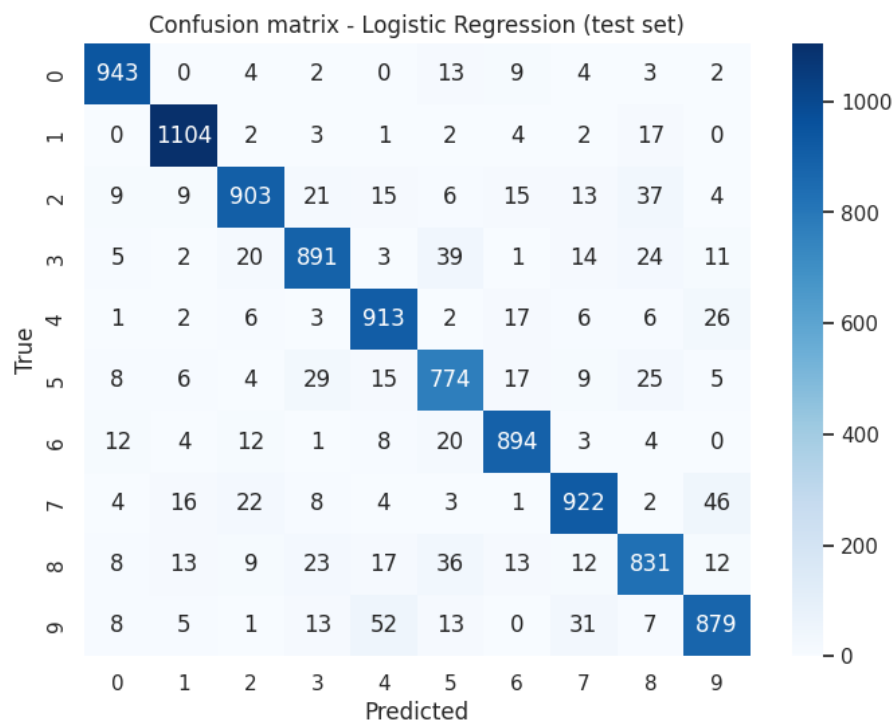
Model	Accuracy (%)
Logistic Regression	89.64 ± 0.83
Logistic (L1)	89.52 ± 0.76
Logistic (L2)	89.64 ± 0.83
SVM (Linear)	90.82 ± 0.47
Decision Tree	75.44 ± 1.10

- SVM achieved the highest accuracy, followed by logistic regression.
- L1/L2 regularization did not significantly improve performance on this dataset.
- Decision trees underperformed due to overfitting.

Classification report (LogReg on test set):				
	precision	recall	f1-score	support
0	0.94	0.96	0.95	980
1	0.95	0.97	0.96	1135
2	0.92	0.88	0.90	1032
3	0.90	0.88	0.89	1010
4	0.89	0.93	0.91	982
5	0.85	0.87	0.86	892
6	0.92	0.93	0.93	958
7	0.91	0.90	0.90	1028
8	0.87	0.85	0.86	974
9	0.89	0.87	0.88	1009
accuracy			0.91	10000
macro avg	0.90	0.90	0.90	10000
weighted avg	0.91	0.91	0.91	10000

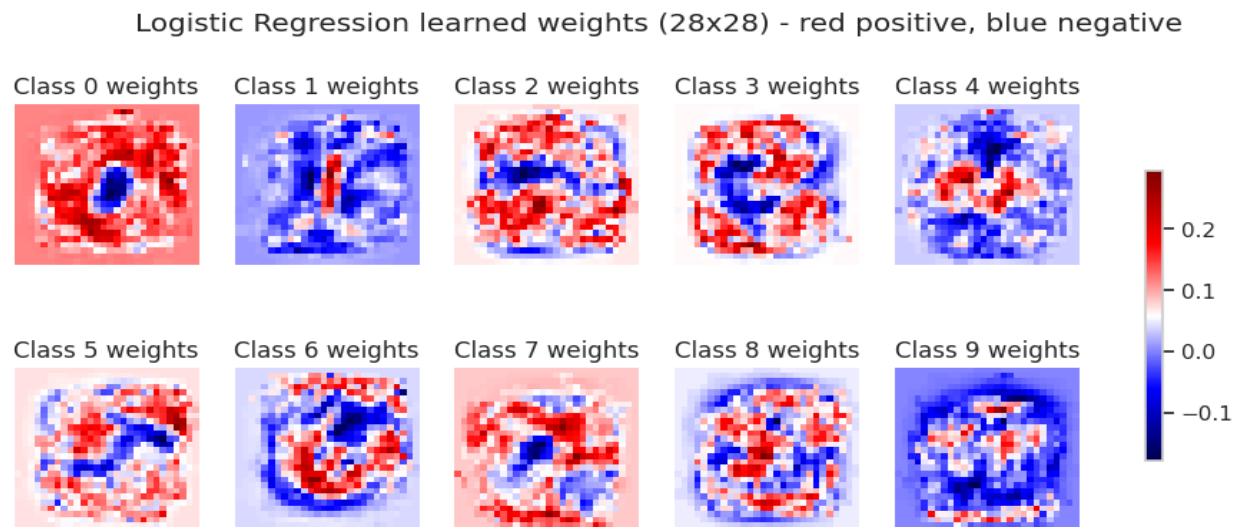
Confusion Matrix (Logistic Regression)

Misclassifications occurred most frequently between visually similar digits (e.g., 3↔5, 4↔9).



Feature Weights Visualization

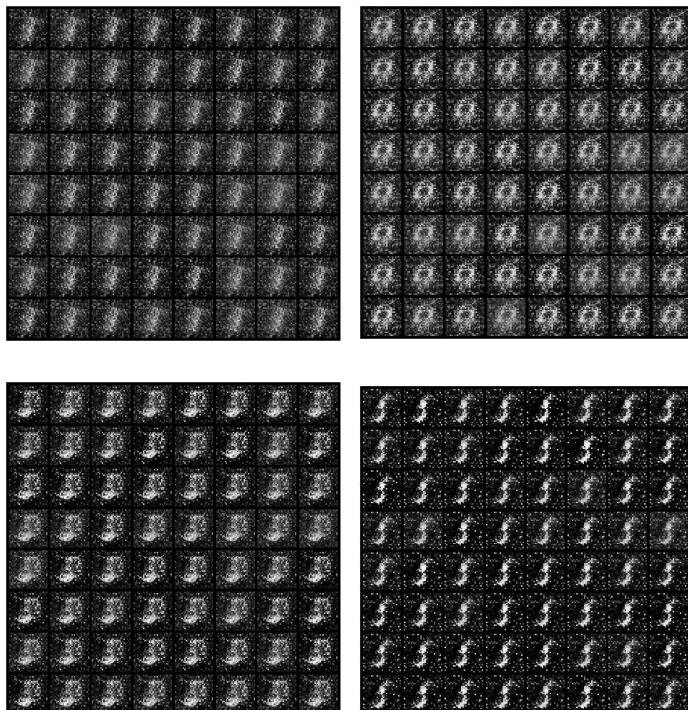
Learned weights per class highlighted stroke patterns (e.g., loops in '8', vertical bars in '1'). Negative weights (blue) often surrounded positive weights (red), suggesting the model learned to suppress irrelevant pixels.



Generative Model (GAN)

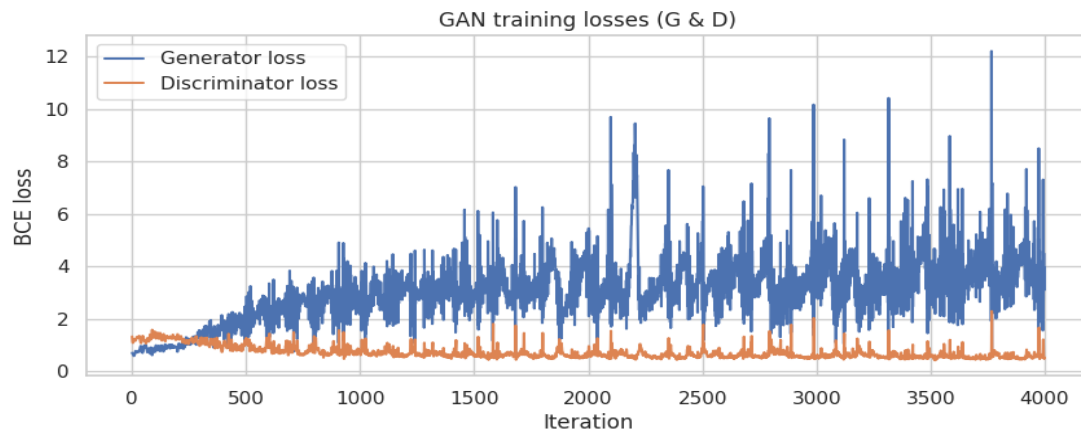
Training Progress

Generated images improved over epochs (see sample grids at steps 500–4000). Early outputs were noisy; later outputs resembled digits but lacked sharpness.



Loss Curves

Generator and discriminator losses oscillated without clear convergence, indicating training instability.

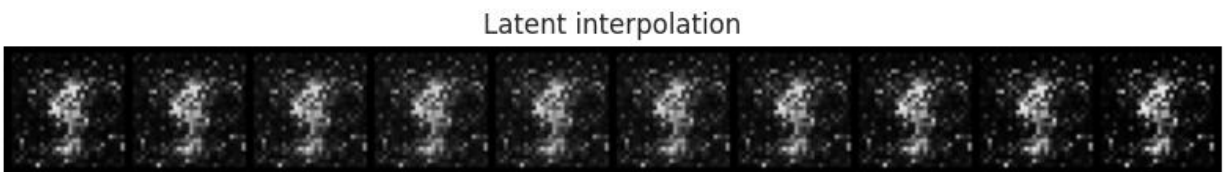


Mode Collapse

When generator capacity was increased (hidden layers [1024,2048,4096]) and learning rate raised ($LR_G=1e-3$), mode collapse occurred: generated samples lacked diversity (e.g., all outputs resembled '1's).

Latent Interpolation

Linear interpolation between noise vectors produced smooth transitions between digit forms, demonstrating continuous latent space.



Synthetic Data Experiment

Logistic regression trained on 2,000 GAN-generated images achieved only **9.74% accuracy** on the real test set, indicating a significant domain gap.

4. Reflection and Analysis

Discriminative vs. Generative Tasks

- Discriminative models are easier to train and evaluate, with clear convergence metrics.
- GANs are unstable due to adversarial dynamics, requiring careful hyperparameter tuning.

Hyperparameter Sensitivity

- GAN training was highly sensitive to architecture and learning rates. Unbalanced capacities led to mode collapse.
- Label smoothing helped stabilize training but did not prevent oscillations.

Literature Insight

As noted in Salimans et al. (“Improved Techniques for Training GANs”, 2016), GAN instability arises from non-convergence, mode collapse, and sensitivity to hyperparameters. Our observations align with these challenges.

Model Debugging

- Discriminative models were easier to debug due to interpretable weights and stable training.
- GANs required iterative experimentation and visual inspection of outputs.

5. Conclusion

Real-World Applications

- **Discriminative Models:** Preferred for classification tasks with labeled data (e.g., spam detection, medical diagnosis).
- **Generative Models:** Useful for data augmentation, art generation, and simulating realistic data where labels are scarce.

Recommendations

- Use discriminative models for well-defined classification problems.
- Employ GANs only when generative tasks are essential, and invest in stabilization techniques (e.g., Wasserstein GAN, gradient penalty).